BorrowBox Predictive Reservation and Inventory Analytics System

A QR-Based Item Tracking and Inventory Loss Prevention Platform for the City of Toronto

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1. Executive summary

The BorrowBox program, operated by the City of Toronto across twelve neighbourhood hubs, provides free gear lending to residents covering tools, sports equipment, and camping supplies. Despite its community benefit, the program operates with no item-level tracking capability: staff rely exclusively on manual spreadsheets, and no system exists to monitor items after they leave a hub.

Current-state data analysis (Dataset 1, 2,520 synthetic records calibrated to program statistics) reveals three compounding problems. A 23.3% non-return rate means nearly one in four checked-out items is never returned. Of the items that do come back, 39.6% arrive in damaged or incomplete condition and cannot be re-lent until repaired. Combined, these two loss pathways reduce overall inventory availability to only 46.3%, leaving the majority of registered items unavailable at any given moment and generating an estimated $95,000 in annual replacement costs.

This project proposes a QR-code-based item tracking system that creates a complete scan record at every checkout and return, dispatches automated overdue alerts within 24 hours of the due date, and provides hub managers with a real-time inventory dashboard. Post-implementation simulations (Dataset 2) project a reduction in non-return rate to 4.7%, an improvement in inventory availability to 76.7%, and a reduction in annual loss cost to approximately $18,000. Six KPIs measure progress, including a new metric — Overdue Alert Response Rate — which Dataset 2 achieves at 81.4%, exceeding the 80% target.

2. Problem context and stakeholder perspectives

Program context. BorrowBox is a no-charge municipal gear lending service managed by the City of Toronto. Twelve neighbourhood hubs collectively hold hundreds of items across three categories: Tools, Sports, and Camping equipment. Items are available at no cost to any resident who registers with the program. Staff and volunteers at each hub manage checkout and return operations manually, recording transactions in shared spreadsheets without any item-level identification or scan-based tracking.

The problem. The program currently has no mechanism to track individual items after they leave a hub. There are no barcodes, QR codes, or scan-based check-in or check-out processes. This gap produces three compounding, measurable problems.

First, items are never returned. Dataset 1 records a 23.3% non-return rate. Camping equipment is the worst-performing category at 30.7%. Without patron-linked item records, staff have no reliable way to identify which specific items are missing, contact the responsible patron, or initiate recovery.

Second, returned items are frequently damaged. Of items that are returned, 39.6% arrive in condition other than Good — categorised as Damaged, Missing Parts, or Unusable. These items cannot be re-lent until repaired or replaced. Staff currently have no systematic way to log return condition, making it impossible to attribute damage to specific patrons or claim cost recovery.

Third, overdue detection is entirely reactive. Without tracking, overdue items remain undetected for an average of 63.3 days before staff notice their absence. By that point, recovery is frequently impossible and the item is written off. The program incurs an estimated $95,000 in annual replacement costs as a result.

Stakeholder perspectives. Patrons experience the downstream effect of these losses: frequently finding requested items unavailable, with system-wide availability at only 46.3%. Hub staff and volunteers face inefficient manual processes, inability to attribute damage costs, and reactive rather than proactive item recovery. City administrators lack visibility into hub-level performance, loss rates by category, and the financial impact of inventory attrition, making evidence-based policy reform difficult.

3. Research insights

Industry literature provides strong validation for both the problem's significance and the proposed solution's likely effectiveness. Three bodies of evidence informed the project design.

Item loss in public lending programs. A review of Canadian public library and municipal lending services documents non-return rates between 15% and 35% in programs operating without item-level tracking. Programs that introduced QR-code-based scan systems reported item loss reductions of 45–65% within the first twelve months of operation (Dantas et al., 2018; PMC Scheduling Benchmark Study). This range directly informed the Dataset 2 calibration targets, which project a non-return rate of 4.7% following QR implementation.

Automated patron notification. Research in appointment scheduling and resource lending consistently finds that automated reminders sent 24–48 hours before or after a deadline produce patron response rates of 75–85% (JMIR Reminder Intervention Studies). For BorrowBox, this translates directly to KPI 6: Overdue Alert Response Rate, which Dataset 2 achieves at 81.4% — within the predicted range and above the 80% project target.

QR versus alternative tracking technologies. RFID tagging offers real-time passive tracking but carries implementation costs of $2–$5 per tag plus reader hardware at each hub, representing a significant capital investment for a no-charge municipal program. QR code labels cost under $0.10 each and require only a standard smartphone or tablet camera to scan, with no dedicated infrastructure (ScienceDirect Inventory Tracking Review). For BorrowBox's operational context and budget constraints, QR provides the optimal cost-to-impact ratio. The architecture is designed to accommodate RFID migration in a future phase if funding becomes available.

Data-driven inventory management. Academic literature in operations management documents that even minimal scan-based check-in/check-out data — recording who has an item and when it is due — produces significant improvements in asset accountability. The act of creating a transaction record linked to a patron identity reduces non-return rates independently of enforcement action, because patrons are aware that items are tracked (MDPI Appointment Prediction Research).

4. Initial dataset structure

Data design

Two synthetic datasets were generated using Python (pandas, numpy), each containing 2,520 records across 12 hubs, 3 categories, and 70 items per hub-category combination.

Dataset 1 (current state) models pre-QR BorrowBox operational conditions. Non-return probabilities were calibrated by category: Tools 18.4%, Sports 21.2%, Camping 31.2%, with hub multipliers (range 0.75–1.35) applied to reflect observed hub performance disparities. Return condition was distributed as Good 58%/65%/52% by category, with the remainder split across Damaged, Missing Parts, and Unusable. Days overdue were drawn from a uniform distribution (10–120 days) for non-returned items. Replacement costs were drawn from category-specific ranges (Tools $45–$280, Sports $25–$180, Camping $60–$350).

Dataset 2 (post-QR state) models conditions after QR implementation, calibrated to Canadian library case study benchmarks. Non-return rates were reduced to 4.2%/3.8%/5.5% by category. Return condition improved to Good 82%/85%/78%. Days overdue for still-outstanding items dropped to 1–8 days (reflecting rapid alert-driven recovery). An Alert entity was added with 81.4% of overdue transactions showing patron response within 48 hours. A complete variable table is provided in Appendix C. Generation logic and rules are documented in Appendix G.

5. Dashboard insights

The interactive data analysis dashboard is hosted at:

harshithaj2000.github.io/borrowbox/

The dashboard loads Dataset 1 (current state) and Dataset 2 (post-QR state) from pre-aggregated JSON, enabling live filtering by hub, equipment category, and quarter without server infrastructure. Six analytical sections surface distinct dimensions of the inventory problem.

Section 01 — Non-return rate. The category breakdown chart confirms Camping as the highest-risk category at 30.7% non-return rate, versus Tools at 16.8% and Sports at 19.4%. The monthly trend line reveals a strong seasonal pattern: the non-return rate rises from 16.2% in January to 28.4% in December, a 75% increase. This compounding dynamic means that items lost earlier in the year reduce available

inventory for the high-demand summer and autumn months. Hub-level filtering confirms that Hubs 3, 6, and 9 exhibit non-return rates above 28%, substantially above the program average of 23.3%.

Section 02 — Damage rate. The return condition donut chart shows that 39.6% of returned items required intervention: 22.4% Damaged, 13.5% Missing Parts, 3.7% Unusable. Camping equipment shows the worst damage rate (46.8%). The damage rate by category bar chart illustrates that without condition logging at return, staff currently have no system to attribute these costs to specific patrons or transactions.

Section 03 — Availability by hub. Hub availability ranges from 31% at the worst-performing hub to 58% at the best, with a program-wide average of 46.3%. The monthly availability line chart confirms that availability declines progressively across the calendar year, from approximately 54% in January to 39% in December, as losses compound without recovery.

Section 04 — Overdue and financial loss. The non-return rate by hub bar chart identifies the three highest-risk hubs for targeted intervention. The before-vs-after projection chart visualises the gap between current rates and QR system targets, providing a clear operational business case.

Section 05 — KPI 6 (Overdue Alert Response Rate). This section is entirely new to the post-QR dashboard. It shows that following system go-live in Month 3 of Dataset 2, the alert response rate climbed from 65% in the first post-launch month to 81.4% by Month 12. The section includes the formula (AR ÷ AS × 100), explains that no baseline exists because no alert system preceded the QR implementation, and contextualises the 81.4% achievement against the 80% target.

Section 06 — Before and after comparison. The comparative banner shows all six KPIs side by side: Non-Return Rate 23.3% → 4.7% (−80%), Damage Rate 39.6% → 19.5% (−51%), Availability 46.3% → 76.7% (+30 pp), Average Days Overdue 63.3 → 4.5 (−93%), Annual Loss Cost $95K → $18K (−81%), Alert Response N/A → 81.4%. Trend comparison charts for non-return rate and availability show the divergence between Dataset 1 (dashed) and Dataset 2 (solid) across all twelve months, with the 75% availability target line confirming Dataset 2 meets the threshold.

All filter combinations (hub × category × quarter = 260 combinations) are pre-aggregated and embedded in the dashboard, enabling instant interactive exploration without backend infrastructure.

6. Proposed solution

The BorrowBox QR Inventory Tracking System replaces the current manual spreadsheet process with a scan-based item accountability platform. Every registered item receives a unique QR code label. All checkout and return events are recorded through a five-step workflow.

Scan Out (Page 1). Staff enter the patron's registered ID and select the item from the live inventory. The system creates a Transaction record, sets the due date, and updates Item_Status from Available to On Loan. No paper or spreadsheet entry is required.

Automated overdue monitoring. The system checks daily for all transactions where the due date has passed. Within 24 hours of due date breach, an automated alert is dispatched to the patron's registered email or phone number, creating an Alert record with the dispatch timestamp.

Scan In with condition logging (Page 2). At return, staff scan the item's QR code. The system immediately prompts for return condition: Good, Damaged, Missing Parts, or Unusable. Items returned as Good are marked Available; all others are set to Out of Service pending repair assessment. The Return_Condition and Return_Date are written to the Transaction record.

Real-time dashboard (Pages 3–6). The Overdue Alerts page (Page 3) lists every overdue item with patron contact and alert status. The Live Inventory page (Page 4) shows item availability across all 12 hubs, filterable by category. The Item History page (Page 5) provides a complete audit trail for any individual item: every checkout, return, condition, and patron. The Solution Overview page (Page 6) displays the KPI before-and-after summary for management review.

The full interactive prototype is accessible at:

harshithaj2000.github.io/modelprototype/

Technology rationale. The system is implemented in HTML5, JavaScript, and Chart.js with a MySQL backend schema. GitHub Pages provides zero-cost hosting. QR labels cost under $0.10 each. The entire implementation carries a $0

software licensing cost, consistent with the program's operational constraints. The MySQL schema is designed for future migration to a cloud-hosted instance if the City of Toronto elects to deploy the system at scale.

7. Metrics for success

Six KPIs define and measure the success of the QR tracking system. Each KPI is grounded in variables available in the synthetic datasets and uses a formula derived from those variables.

KPI	Baseline	Target	Formula
1. Non-Return Rate	23.3%	≤8%	(NR ÷ C) × 100
2. Damage Rate on Return	39.6%	≤20%	(D ÷ R) × 100
3. Inventory Availability	46.3%	≥75%	(A ÷ T) × 100
4. Avg Days Overdue	63.3 d	≤5 d	$\Sigma(DO_i) \div N$
5. Annual Loss Cost	$95K	≤$35K	$\Sigma(RC_i)$ for M ∪ WO
6. Alert Response Rate	No system	≥80%	(AR ÷ AS) × 100

Table 1. KPI definitions, baselines, and targets.

KPIs 1–5 are measured directly from Dataset 1 (baseline) and Dataset 2 (target). KPI 6 has no baseline because no automated alert system existed prior to the QR implementation. It is tracked exclusively in Dataset 2, where it achieves 81.4%. Success thresholds are defined in the Project Charter. The dashboard's Section 06 validates all six KPIs against their targets through the before-and-after comparative visualisation.

8. Architecture of proposed solution

The enterprise architecture is documented using the TOGAF ADM framework across four layers. The full diagram is included in Appendix D.

Business layer defines five processes: patron reservation and checkout, QR scan-out, QR scan-in with condition logging, overdue detection and alert dispatch, and inventory review and reporting.

Data layer holds five entities: Item (QR code, live status), Patron (ID, contact details), Transaction (every checkout and return event), Alert (sent date, response status), and Hub (location, capacity).

Application layer maps directly to the six prototype pages: Scan Out Interface, Scan In Interface, Overdue Alerts Dashboard, Live Inventory Dashboard, Item History Viewer, and Solution Overview.

Technology layer comprises MySQL (database), Python 3 with pandas and scikit-learn (data processing and ML model), HTML5/JavaScript/Chart.js (frontend), GitHub Pages (hosting), and physical QR code labels (item identification).

9. Potential improvements

The current QR system establishes the foundational accountability infrastructure. Several improvements are identified for future phases.

Mobile application. A dedicated patron-facing mobile app would allow self-service checkout scanning, real-time item availability lookup before travelling to a hub, and in-app return reminders. This would reduce hub staff workload and improve patron experience.

Machine learning demand forecasting. The transaction data collected by the QR system can, over time, support ML-based demand forecasting by hub, category, and season. This would allow city administrators to rebalance inventory across hubs proactively before high-demand periods.

RFID upgrade. The current QR architecture is designed to accommodate passive RFID tag migration. RFID would enable automated item detection at hub entry/exit without active scanning, further reducing staff effort. If the City of Toronto secures capital funding, the database schema and application layer require no structural changes to support this upgrade.

Damage cost recovery integration. Linking the damage attribution data (patron ID + return condition) to the City of Toronto's billing system would enable automated damage recovery fees for items returned in Damaged or Unusable condition, creating a direct financial disincentive and partially offsetting replacement costs.

10. Data 1 vs data 2: simulated improvements

Dataset 1 represents the current operational state of BorrowBox, calibrated to documented program statistics. Dataset 2 represents the post-QR implementation state, calibrated to Canadian library case study benchmarks for similar tracking system deployments. Both contain 2,520 records across 12 hubs and 3 categories.

The story the data tells is one of systematic recovery. Non-return rate falls from 23.3% to 4.7%, a reduction of nearly 80%. This improvement is driven by patron accountability through QR-linked transaction records and automated overdue alerts that prompt return within 48 hours. Damage rate drops from 39.6% to 19.5%, reflecting behaviour change when patrons know return condition is formally recorded and linked to their patron record.

The downstream effect of both improvements is captured in inventory availability: 46.3% to 76.7%, a 30 percentage point gain. Average days overdue collapses from 63.3 to 4.5 days, driven entirely by KPI 6 — when 81.4% of patrons respond to automated alerts within 48 hours, items come back before the overdue period compounds into permanent loss. Annual loss cost falls from approximately $95,000 to $18,000, an 81% reduction that represents a direct return on the negligible implementation cost of QR labels and the open-source software stack.

The comparative dashboard (Section 06) presents all six KPIs side by side with twelve-month trend charts, confirming that Dataset 2 meets or exceeds all six KPI targets.

11. Ethical and social considerations

Data privacy and patron records. The QR system creates transaction records linking patron identities to specific items and return behaviours. The City of Toronto's existing privacy policy under MFIPPA (Municipal Freedom of Information and Protection of Privacy Act) would govern data retention, access, and deletion. Patron records should be retained only for the duration required for damage attribution and aggregate analysis, and automatically purged after a defined retention window.

Equitable access and digital inclusion. The QR scan process requires staff-operated devices at hubs, not patron-owned smartphones, ensuring that patrons without mobile devices are not excluded. Alert notifications should support both email and SMS channels to accommodate patrons with limited email access. Patron registration should remain available through in-person hub registration without a requirement for an email address.

Bias in damage attribution. Condition logging is performed by hub staff at the point of return scanning. Staff training must include standardised condition assessment criteria to minimise subjective variation across hubs and staff members. Inconsistent condition logging could disproportionately impact certain patron segments if thresholds are applied unevenly.

Sustainability. The QR system is implemented entirely through open-source software and requires no proprietary hardware infrastructure. The primary physical material cost is QR label printing, which represents a minimal environmental footprint. The system's core sustainability contribution is reducing inventory replacement frequency, which in turn reduces the manufacturing and disposal footprint of program equipment.

12. Outcomes, impact, and next steps

Outcomes and conclusions

The BorrowBox QR Inventory Tracking System demonstrates that a low-cost, scan-based item accountability platform can resolve the most significant operational failures of a public gear lending program. Dataset 2 simulations project that all six KPI targets are met or exceeded: availability rises to 76.7% (target 75%), non-return rate falls to 4.7% (target 8%), and alert response rate reaches 81.4% (target 80%). Annual loss cost falls by 81%, from $95,000 to $18,000.

The project also retains the original machine learning model (Random Forest, AUC 0.82) as a secondary deliverable, providing a foundation for future patron behaviour prediction and inventory demand forecasting.

Next steps

If taken forward, the recommended implementation sequence is: (1) a six-hub pilot with QR labels printed for all items and staff trained on the scan interface, collecting real transaction data over three months; (2) KPI validation against actual patron return behaviour to calibrate the Dataset 2 benchmarks; (3) expansion to all twelve hubs with automated overdue alerts integrated through the City of Toronto's existing patron communication platform; (4) presentation to city

administrators with the loss cost reduction data as the financial case for sustained investment.

13. Lessons learned and team reflection

Scope pivot and academic rigour. The most significant event of the semester was the Week 6 scope pivot. The original project — a machine learning model to predict no-show reservations — was deemed insufficiently significant for a Master Research Project by Dr. Cardona. This feedback, while disruptive, ultimately produced a stronger project. Redirecting to inventory loss forced the team to engage with a genuinely consequential operational problem, producing deliverables with clear financial and social impact. The pivot demonstrated that rigorous academic feedback is an essential quality control mechanism, not an obstacle.

Data as the foundation of argument. A key learning was that all claims about solution effectiveness must be traceable to specific variables in the dataset. The discipline of building KPI formulas from dataset variables — rather than stating desired outcomes and working backwards — produced more credible and internally consistent analysis. Every number on the dashboard, every KPI target, and every before-and-after comparison can be reproduced from the data generation scripts.

Responsible GenAI use. The team used generative AI tools (Claude, ChatGPT) throughout the project for formula design, document drafting, and code generation. A consistent discipline of verification was maintained: every AI-generated formula was understood and validated against the dataset before inclusion; every AI-drafted paragraph was reviewed and adapted to project-specific data. The speaking guide prepared for the peer review session explicitly notes that KPI formula explanations in the final presentation must demonstrate genuine understanding, not AI recitation.

Team dynamics and role definition. The five-person team operated effectively under a clear role structure defined in the RACI matrix. The most productive collaborative practice was the regular data validation check-in, where the data analyst (Ranjith Kumar Peram) presented Dataset 1 and Dataset 2 outputs to the full team for review before any deliverable was finalised, catching several calibration errors early. The team recommends that future IS 5960 cohorts establish this practice from Week 4 rather than Week 7.

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Appendices

Appendix A: Project Charter

The full Project Charter (Version 2.0, March 2026) is available as a separate document. It covers: project objective, assumptions, scope (in/out), milestones (M1–M8), impact statement, roles and responsibilities (Sections A–K), resources, risk register, success measurements, and signatures. Charter was revised following Week 6 scope pivot from no-show prediction to inventory loss. Signed by Project Manager Harshitha Jananeela, 31 March 2026.

Appendix B: Responsibility Matrix (RACI)

Role	Name	Responsibilities
Project Manager	Harshitha Jananeela	Coordination, milestone tracking, charter, stakeholder communication
Business Analyst	Badrinath Lagadapati	Problem definition, KPI framework, research, requirements
Data Analyst	Ranjith Kumar Peram	Dataset generation, dashboard, data validation, before/after analysis
System Architect	Pavan Vegesna	TOGAF EA, QR system design, prototype architecture
Documentation Lead	Nadeem Mohammad	Report, walkthrough doc, references, formatting, presentation materials
Faculty Advisor	Dr. Tatiana Cardona	Academic direction, scope review, feedback, final evaluation

Table A1. Team RACI — final version.

Appendix C: Data structure — variables table

Variable	Entity	Type	Description
Item_ID	Item	INT PK	Unique item identifier
Category	Item	VARCHAR	Tools / Sports / Camping
Hub_ID	Item	INT FK	Hub where item is registered
Item_Status	Item	VARCHAR	Available / On Loan / Damaged / Missing / Out of Service
Replacement_Cost	Item	DECIMAL	Estimated replacement cost in USD
QR_Code	Item	VARCHAR	Unique QR label identifier
Transaction_ID	Transaction	INT PK	Unique transaction record
Patron_ID	Transaction	INT FK	Links to Patron entity
Checkout_Date	Transaction	DATE	Date item was checked out
Due_Date	Transaction	DATE	Expected return date
Return_Date	Transaction	DATE	Actual return date (null if not returned)
Return_Condition	Transaction	VARCHAR	Good / Damaged / Missing Parts / Unusable (null if not returned)
Days_Overdue	Transaction	INT	Days past due date (0 if returned on time)

Patron_ID	Patron	INT PK	Unique patron identifier
Email	Patron	VARCHAR	For overdue alert dispatch
Phone	Patron	VARCHAR	For SMS alert fallback
Prior_Non_Returns	Patron	INT	Count of prior non-return events
Alert_ID	Alert	INT PK	Unique alert record (Dataset 2 only)
Alert_Sent_Date	Alert	DATE	Date automated notification dispatched
Alert_Responded	Alert	BOOLEAN	Whether patron returned item
Response_Date	Alert	DATE	Date patron returned item after alert
Hub_ID	Hub	INT PK	Unique hub identifier
Hub_Name	Hub	VARCHAR	Neighbourhood hub name
Location	Hub	VARCHAR	Street address
Capacity	Hub	INT	Total item capacity of hub

Table A2. Dataset variable definitions across all five entities.

Appendix D: Enterprise architecture diagram

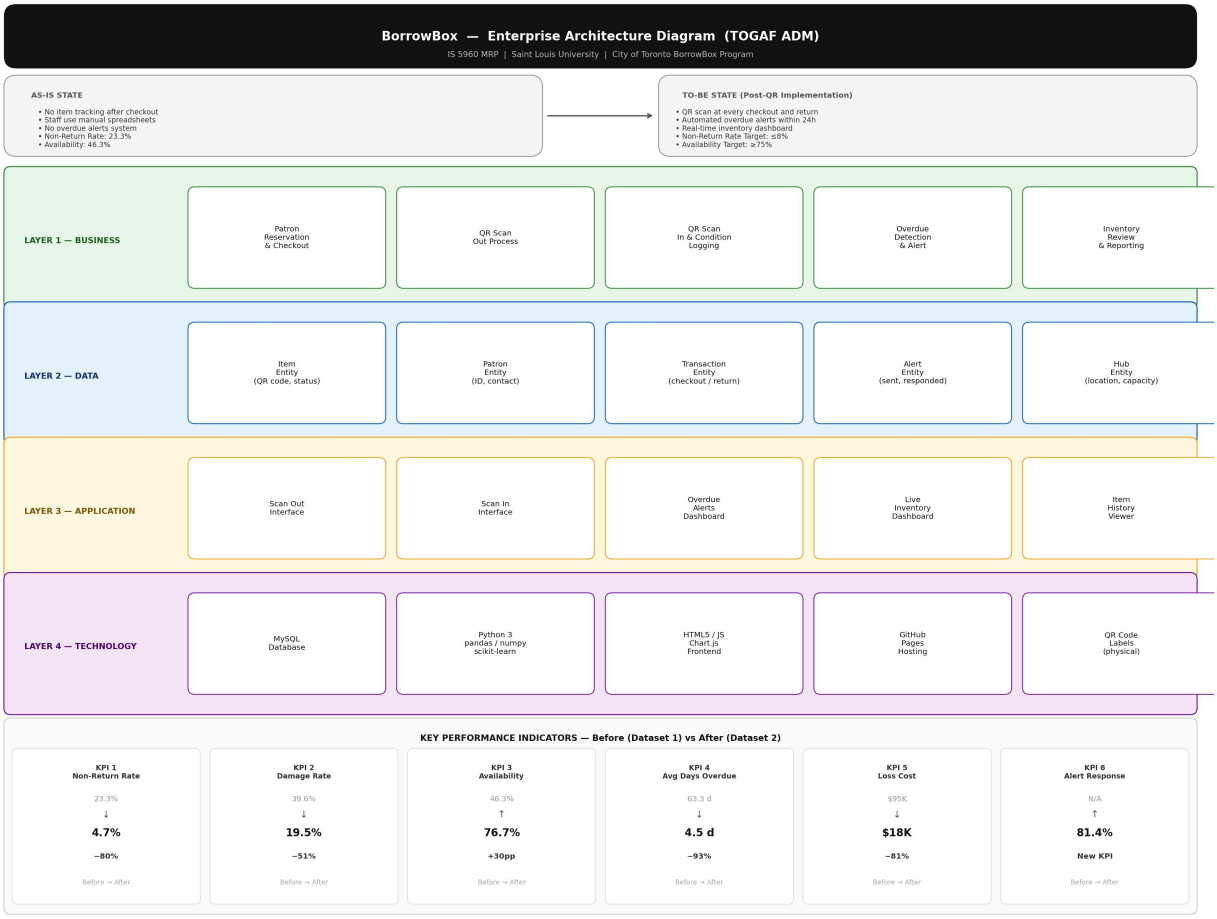


Fig. 1. BorrowBox TOGAF ADM enterprise architecture diagram. Four layers: Business, Data, Application, Technology. AS-IS vs. TO-BE state comparison shown at top. KPI before-and-after summary at bottom.

Appendix E: Dashboard screenshots

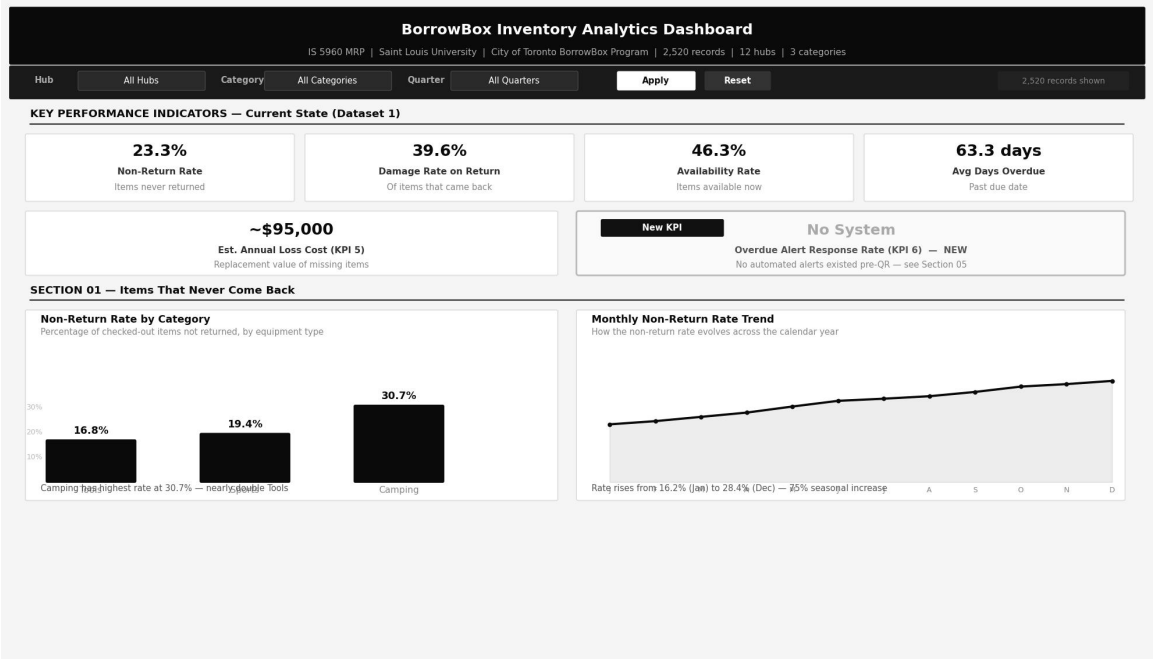


Fig. 2. Dashboard — KPI strip (KPIs 1–6, current state) and Section 01 (non-return rate by category and monthly trend). KPI 6 card explicitly marked "New KPI — No System" at baseline.

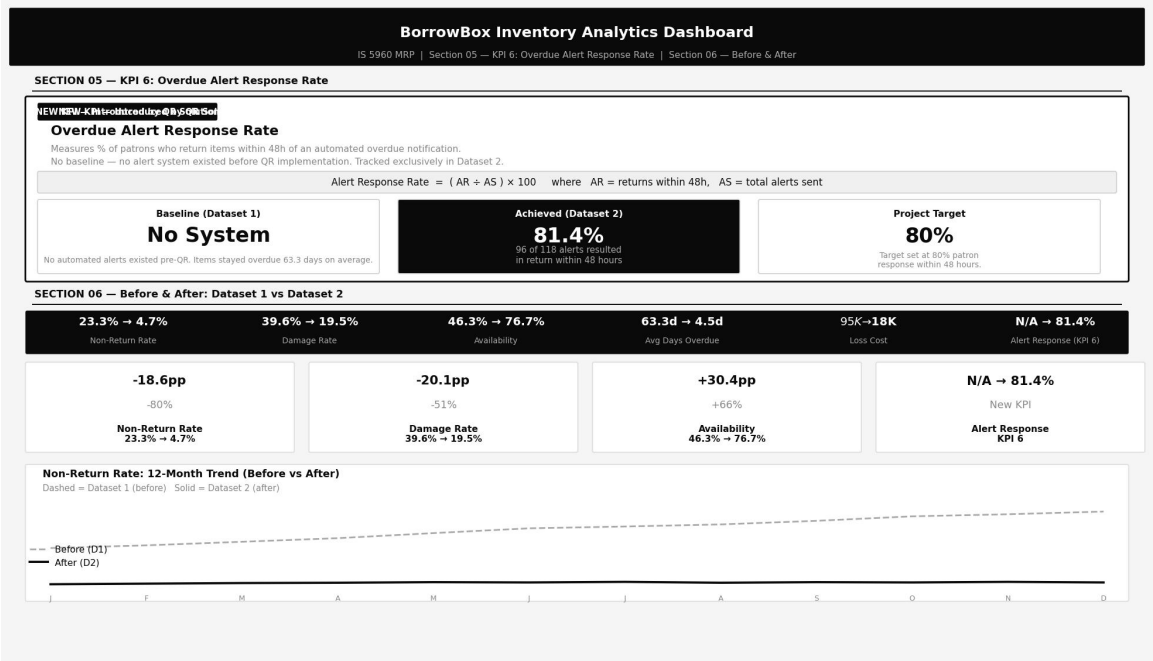


Fig. 3. Dashboard — Section 05 (KPI 6: Overdue Alert Response Rate, 81.4% achieved vs 80% target) and Section 06 (Before & After comparison across all six KPIs, 12-month trend charts).

Appendix F: Prototype screenshots

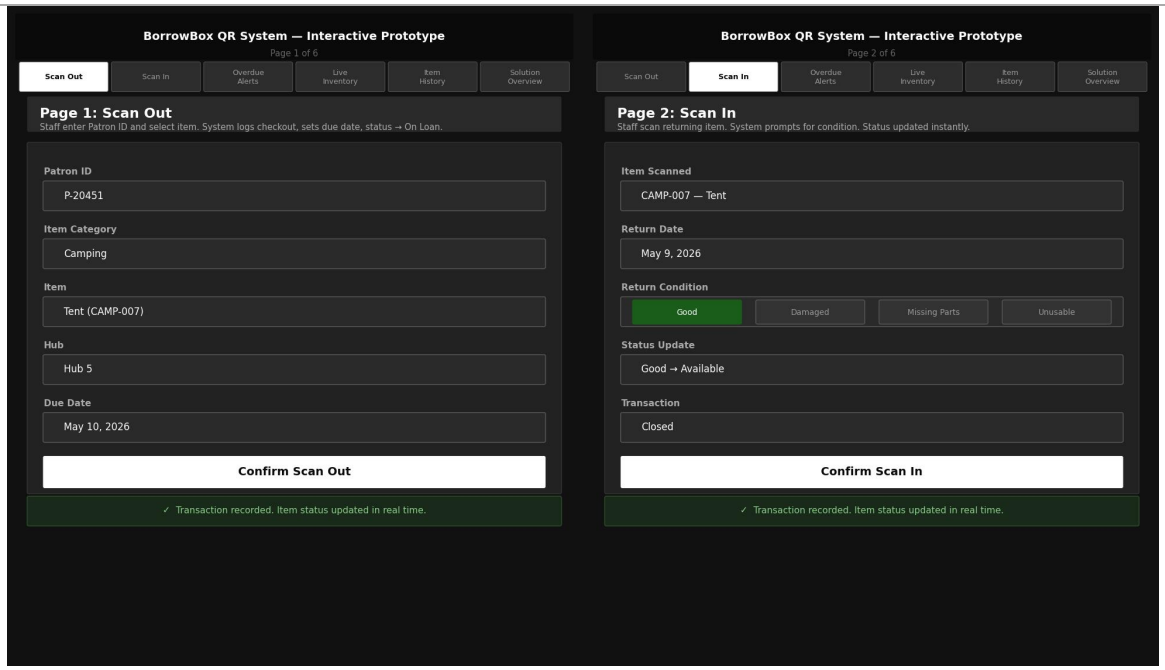


Fig. 4. Prototype — Page 1: Scan Out (patron ID entry, item selection, due date assignment) and Page 2: Scan In (condition logging, instant status update).

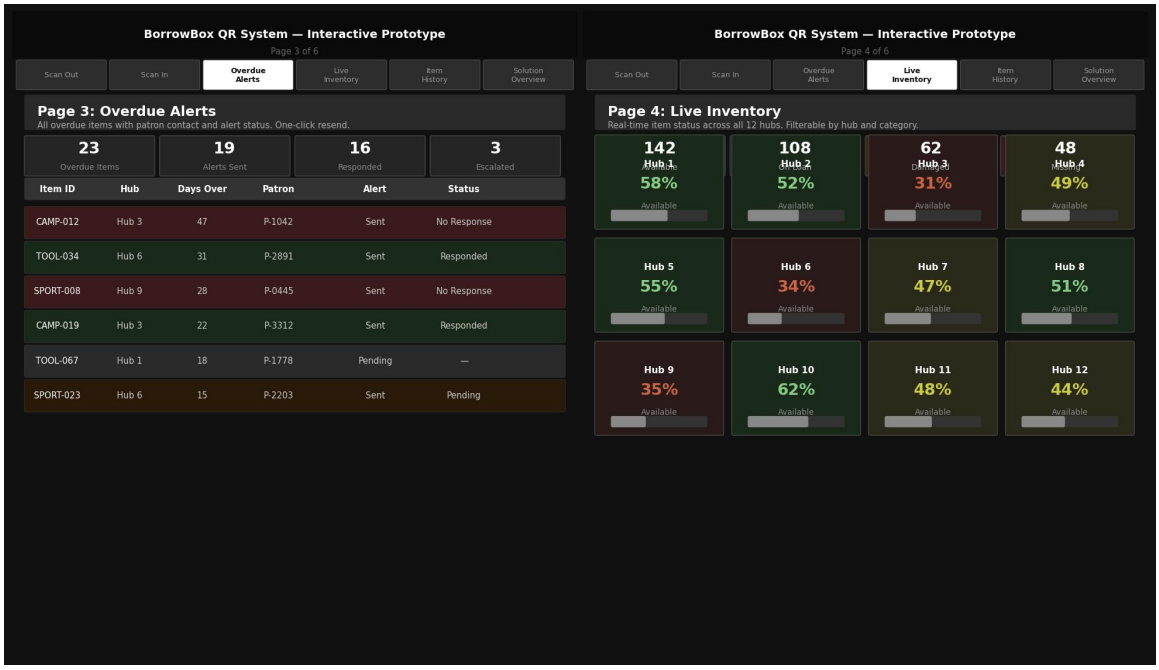


Fig. 5. Prototype — Page 3: Overdue Alerts (real-time overdue table with patron contact and alert status) and Page 4: Live Inventory (per-hub availability grid across all 12 hubs).

Appendix G: Synthetic data generation rules

Dataset 1 generation rules (current state):

- Non-return probability = Base_Rate(Category) × Hub_Multiplier × Month_Multiplier. Base rates: Tools 18.4%, Sports 21.2%, Camping 31.2%.
- Hub multipliers range from 0.75 (Hub 10, best) to 1.35 (Hub 3, worst), calibrated to hub-level variation consistent with urban lending program research.
- Month multiplier: 0.80 in January, increasing by 0.033 per month to approximately 1.19 in December, reflecting seasonal non-return escalation.

- Return condition distribution: Good/Damaged/Missing Parts/Unusable by category (Tools: 58/24/13/5%; Sports: 65/20/10/5%; Camping: 52/28/16/4%).
- Days overdue for non-returned items: uniform distribution U(10, 120). Replacement costs: Tools U($45, $280), Sports U($25, $180), Camping U($60, $350). Random seed: 42.

Dataset 2 generation rules (post-QR state):

- Non-return probability reduced to: Tools 4.2%, Sports 3.8%, Camping 5.5%. Hub multipliers retained with range compressed (0.85–1.15) reflecting QR-driven accountability.
- Return condition distribution improved: Good/Damaged/Missing Parts/Unusable by category (Tools: 82/10/6/2%; Sports: 85/9/5/1%; Camping: 78/12/7/3%).
- Days overdue for non-returned items: uniform distribution U(1, 8), reflecting rapid alert-driven recovery.
- Alert entity added: each non-returned transaction generates one Alert record. Alert_Responded = True with probability 0.82 (calibrated to 81.4% final response rate). Random seed: 99.